Indonesia's path to sustainable aviation fuel: Evaluating feedstock potential from agricultural residue

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Highlights

- 171.8 Mt/year of agricultural residues available for SAF feedstock in Indonesia.

- Palm, rice, and corn residues suit thermochemical and biochemical valorization.

- Pyrolysis, HTL, gasification, and AtJ pathways evaluated for local biomass.

- Decentralized pre-processing hubs proposed to mitigate high logistic costs.

- Residue-based SAF offers up to 94 % GHG reduction vs. fossil jet fuel.

Abstract

Sustainable Aviation Fuel (SAF) is an important pathway for decarbonizing the aviation sector. This supports both global climate goals and Indonesia's net-zero targets. Indonesia's agricultural residues—from palm oil, rice, and corn—have big potential as lignocellulosic feedstocks. These residues are different from primary crops. They do not compete with food or the biodiesel sector. This review looks at if it is feasible to turn these residues into SAF. The methods include pyrolysis, hydrothermal liquefaction (HTL),

gasification, and fermentation. Techno-economic analyses show that thermochemical routes give better greenhouse gas (GHG) emission reductions (up to 94 %). Production costs, however, are still high ($1.80–$3.50/L) compared to normal jet fuel. There are problems holding back residue-based SAF use in Indonesia. There is a lack of fiscal incentives, undeveloped infrastructure and, on top of that, the market is a monopoly. The study points to one way to fix this: a need for decentralized pre-processing hubs to cut logistics costs. Also, urgent policy reforms are needed, like price gap funding and tax incentives, to make investment less risky. Strategic collaboration between academia, industry, and government is important to make Indonesia's residual potential a commercial reality.

Introduction

Sustainable Aviation Fuel (SAF) is progressively becoming a focal point, owing to its contribution as an essential component of the industry's plans to decrease its impact on the environment [1]. The global move towards using sustainable options such as SAF is visible in the rules that nations such as Indonesia are setting to achieve higher levels of use of SAF [2]. Businesses are looking for suppliers that can support the growth of biorefinery plants, such as PT Pertamina from Indonesia, to supply the increasing SAF demand that depends on aviation industry net-zero initiatives [1].

Traditional aviation fuels derived from fossil sources are major contributors to greenhouse gas (GHG) emissions, which drive climate change. However, SAF is manufactured from renewable sources, including biomass, agricultural residuals, and waste, and hence is a better option than petrol [[3], [4], [5]]. The application of SAF can save as much as 93 % of the life-cycle greenhouse gas emissions compared to conventional Jet A fuel. This is achieved with the help of fuel feedstocks, which undergo the process of CO_2 sequestration during their growth phase and thus can compensate for the emissions that are generated during the use of the fuel [4,6]. In addition, it was found that SAF can be installed in aircraft engines and other related structures without any alteration [7]; thus, it can be easily integrated into the market.

SAF implementation is necessary for the reduction of emissions through practices such as those legislated by the International Civil Aviation Organisation (ICAO) through the Carbon Offsetting and Reduction Scheme for International Aviation (CORSIA) [8]. For the aviation industry, as it grows, incorporation of SAF in the fuel supply chain is still accepted as an effective tool for the sustainable development of aviation business and the negative effects of the industry on climate change [9].

Based on available natural resources and a suitable geographical location, Indonesia is expected to occupy a significant position in the global SAF market [10]. Indonesia is the world's largest producer of palm oil; thus, it produces large amounts of agricultural residues such as empty fruit bunches and palm kernel shells, which can be utilized as feedstock for SAFs production [[11], [12], [13]]. In addition, a vast number of forests are present in Indonesia, which forms a broad base of the forestry industry and can also provide a great amount of co-products of forestry, such as wood chips and sawdust, which can also be used as feedstock for SAF production [14].

Increases in travel both domestically and internationally through Indonesia, the growing aviation industry, therefore, calls for a search for environmentally sound solutions for aviation emissions. Nowadays, the country demonstrates its willingness to decrease carbon dioxide emissions through its national policies for the utilization of renewable energy and sustainable development [15]. Indonesia is interested in bioenergy and has introduced incentives for biofuel plants, thereby supporting the development of SAF projects [10]. Moreover, Indonesia is located in Southeast Asia, which makes it strategically located to distribute SAF not only to Southeast Asia, but also to other relevant regions in the world. Thus, taking into account the quantitative and qualitative potential of Indonesia and given the country's policy measures, it can be stated that Indonesia can and should play a crucial role in the global process of sustainable development in the aviation industry.

This study aims to assess the feasibility of Indonesia as a source of SAF production by analyzing feedstock availability, both technologically and economically. This work provides a detailed evaluation of the available feedstocks in Indonesia, focusing on agricultural residues as potential feed for SAF production. Furthermore, it aims to determine the economic viability of converting such feedstocks to SAF, the primary conversion processes, and the market situation.

Types of SAF

SAF can be further divided into several types depending on the feedstock and the production processes involved [3]. There are two major sources: biofuel and synthetic fuel [5], as indicated in Fig. 1. Biofuels can be produced from a wide variety of feedstocks ranging from agricultural residues to forest residues, municipal solid waste, and energy crops [3,5]. On the other hand, synthetic fuels produced from biomass and residual waste gasification offer another route towards reducing emissions

Availability and potential of feedstock from agricultural residue for SAF production in Indonesia

Indonesia, with its vast and diverse natural resources, has a significant potential for the production of SAF. The country's unique geographical and agricultural landscapes offer a wide array of feedstock options that can be harnessed for SAF development. This section explores three primary sources of agricultural residue potential as feedstock in Indonesia: palm oil residues, forestry residues, and industrial waste (Table 2).

Each of these sources represents a valuable opportunity to convert

Conversion Process Routes and upgrading technologies

Agricultural residues such as palm oil mill residues, rice residues, and corn residues are promising feedstocks for SAF production. These residues can be transformed into bio-oil, biocrude, or other intermediates through advanced conversion technologies, which are then upgraded to SAF, as shown in Fig. 2. Each residue type has unique characteristics that influence the selection of suitable technologies, economic viability, and fuel yields.

The conversion of agricultural residues into SAF follows

Conclusion and future outlook

Indonesia possesses an immense opportunity to decarbonize the global aviation sector by leveraging its abundant agricultural residues; specifically, palm oil mill wastes (EFB, PKS), rice husks, and corn cobs. Unlike first-generation biofuels that compete with food supply, these lignocellulosic feedstocks offer a sustainable pathway with superior greenhouse gas reduction potential. However, converting this potential into a commercial reality requires shifting focus from assessment to

CRediT authorship contribution statement

Afdal Adha: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Conceptualization. **Muhammad Ilham Adhynugraha:** Writing – review & editing, Visualization, Investigation. **Fithri Nur Purnamastuti:** Writing – review & editing, Visualization, Investigation. **Fadli Cahya Megawanto:** Writing – review & editing, Visualization, Investigation. **Budiyanto:** Writing – review & editing, Visualization, Investigation. **Chairunnisa:** Writing – review & editing,

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Conflict of interest

None declared.

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